

Machine Learning for Imaging – Coursework Report

Mammographic density prediction

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1 Part A: Choose a sensible data augmentation pipeline

To improve generalisation in breast density prediction, we implemented a data augmentation pipeline that mimics realistic variability in mammography acquisition while preserving the global tissue patterns that define BI-RADS density. We separated augmentations into photometric and geometric transformations using `torchvision.transforms.v2`.

For photometric augmentation, we applied mild intensity-domain perturbations to account for scanner- and preprocessing-related differences: `RandomAutocontrast` ($p=0.3$) to vary global contrast, `RandomEqualize` ($p=0.1$) to occasionally alter intensity distributions, `GaussianNoise` ($\sigma=0.03$) to simulate sensor/digitisation noise, and `GaussianBlur` applied with low probability ($p=0.15$) to model small sharpness variations (e.g., motion or reconstruction differences). We also included a random gamma adjustment (via `adjust_gamma`) to capture tone-curve and exposure variability while maintaining anatomical structure.

For geometric augmentation, we used `RandomAffine` with conservative parameters (rotation $\pm 8^\circ$, translation up to 6%, and scale 0.95–1.05) to model plausible positioning differences such as slight tilt, centring shifts, and minor magnification changes. `RandomHorizontalFlip` ($p=0.5$) was included since density labels are invariant to laterality. We avoided aggressive cropping, perspective warps, and elastic deformations as these can remove breast tissue or introduce unrealistic distortions that may change density cues.

2 Part B: Implement a sensible model for image classification

We evaluated four architectures for breast density classification: a two-layer MLP baseline and three convolutional neural networks (CNNs) from `torchvision.models`: ResNet-18, DenseNet-121, and EfficientNet-B0. All CNNs were initialised with ImageNet-pretrained weights. Models were first trained for 6 epochs and evaluated using validation AUROC (Table 1).

Model	Validation AUROC (6 epochs)
Baseline MLP	0.8180
EfficientNet-B0	0.8719
DenseNet-121	0.9162
ResNet-18	0.9185

Table 1: Validation AUROC after 6 epochs.

The MLP performed substantially worse than the CNNs, confirming that convolutional architectures are necessary to capture the spatial structure of mammographic images. Among CNNs, ResNet-18 achieved the highest validation AUROC after 6 epochs (0.9185), closely followed by DenseNet-121 (0.9162), while EfficientNet-B0 underperformed.

For all CNNs, the best checkpoints occurred within the first few epochs (ResNet-18: epoch 1; DenseNet-121: epoch 0; EfficientNet-B0: epoch 4), indicating rapid convergence and effective transfer of ImageNet-pretrained features.

Given the small performance gap between ResNet-18 and DenseNet-121, we trained both models for 10 epochs. ResNet-18 did not surpass its earlier performance (best AUROC 0.9155 at epoch 6). DenseNet-121 achieved a higher peak validation AUROC of 0.9288; however, the best checkpoint again occurred at epoch 0 (step 83). As shown in Figure 1, validation performance fluctuates for both models over training. While DenseNet-121 does not exhibit perfectly stable behaviour, it consistently maintains higher validation AUROC than ResNet-18, which shows greater variability and a pronounced late-epoch performance drop. These dynamics suggest that extended training does not reliably improve generalisation and that the pretrained DenseNet representation is already well-suited to the task.

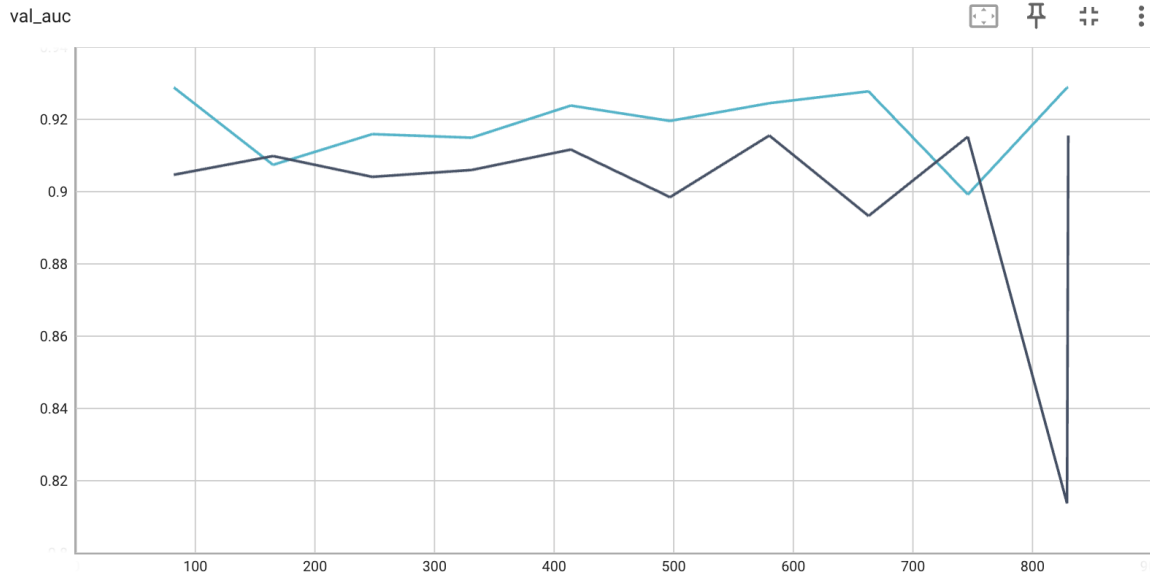


Figure 1: Validation AUROC versus training step for ResNet-18 (dark) and DenseNet-121 (light blue) over 10 epochs. The x-axis shows training steps (mini-batch updates) and the y-axis shows validation AUROC.

Overall, convolutional architectures significantly outperform the MLP baseline. DenseNet-121 achieved the highest validation AUROC and more stable training behaviour, and was therefore selected as the final model. Test AUROC for the selected DenseNet-121 checkpoint was 0.9143.

3 Part C: Conduct a subgroup performance analysis

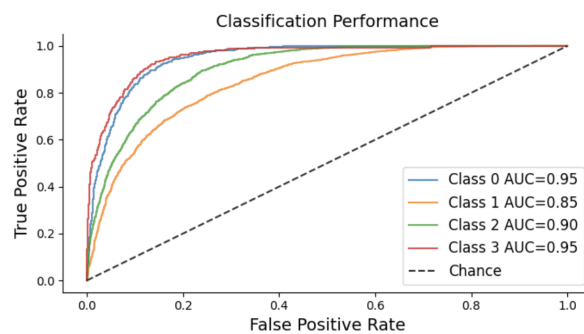


Figure 2: ROC curves for the four BI-RADS density classes. Classes 0–3 correspond to BI-RADS A–D.

Figure 2 shows the ROC curves for the four BI-RADS density classes on the full test set. The model demonstrates strong discrimination overall, with AUC values ranging from 0.85 to 0.95. Performance

was highest for Classes A and D (almost entirely fatty and extremely dense) and lowest for Class B.

To evaluate robustness and fairness, subgroup performance was analysed across age, implant status, imaging view, scanner ID, and laterality using macro-AUC, summarized in Table 2.

Attribute	Subgroup	Macro AUC
Age group	<40	0.865
	40–50	0.910
	50–60	0.904
	60–70	0.920
	70+	0.931
Implant	no	0.921
	yes	0.784
Laterality	Left	0.913
	Right	0.915
Machine ID	49	0.916
	93	0.904
	170	0.941
	210	0.894
	216	0.900
View	CC	0.913
	MLO	0.916

Table 2: Macro-AUC across metadata subgroups.

Age. As shown in Figure 3, performance increased with age, with the lowest macro-AUC observed in patients under 40 (0.865) and the highest in patients over 70 (0.931). This indicates reduced model performance in younger patients.

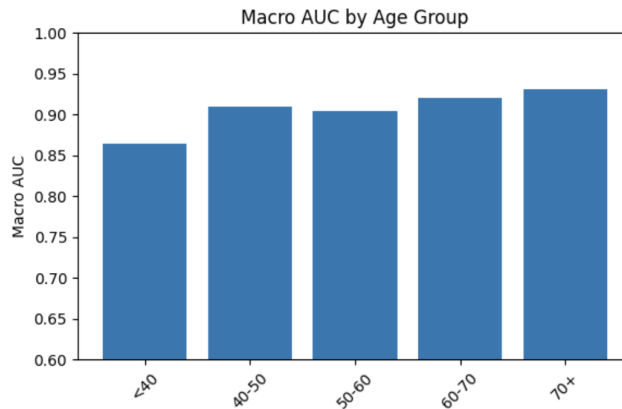


Figure 3: Macro-AUC stratified by age group.

Implants. Performance was substantially lower in cases with implants (0.784 vs 0.921), suggesting reduced generalisation in the presence of implants.

View. Similar performance between CC and MLO projections indicates robustness to projection angle.

Machine ID. Moderate variation across scanners (0.894–0.941) suggests mild sensitivity to acquisition differences, though performance remained stable overall.

Laterality. Nearly identical performance between left and right breasts indicates no observable laterality bias.

Overall, the most pronounced effect was age-dependent variation, with lower performance in younger patients and reduced generalisation in implant cases.

4 Part D: Model inspection

To analyse the learned representation, we extracted 1024-dimensional embeddings from the penultimate layer of DenseNet-121 by replacing the final classifier with an identity mapping and forwarding the test set.

Figure 4 shows a 2-D PCA projection coloured by BI-RADS density. The embeddings form a continuous, ordered structure, with a clear progression from Class A to D along the first principal component. Adjacent classes overlap, while extreme classes are more separated, reflecting the ordinal nature of density grading.

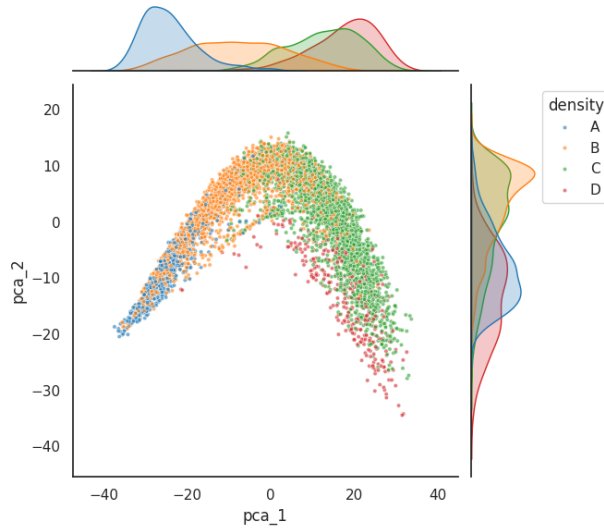


Figure 4: 2-D PCA projection of DenseNet-121 embeddings, coloured by BI-RADS density (A–D).

Colouring embeddings by age (Figure 5) shows no distinct clustering or directional structure, indicating that age does not define the embedding geometry, despite the modest age-dependent AUROC differences observed in Part C.

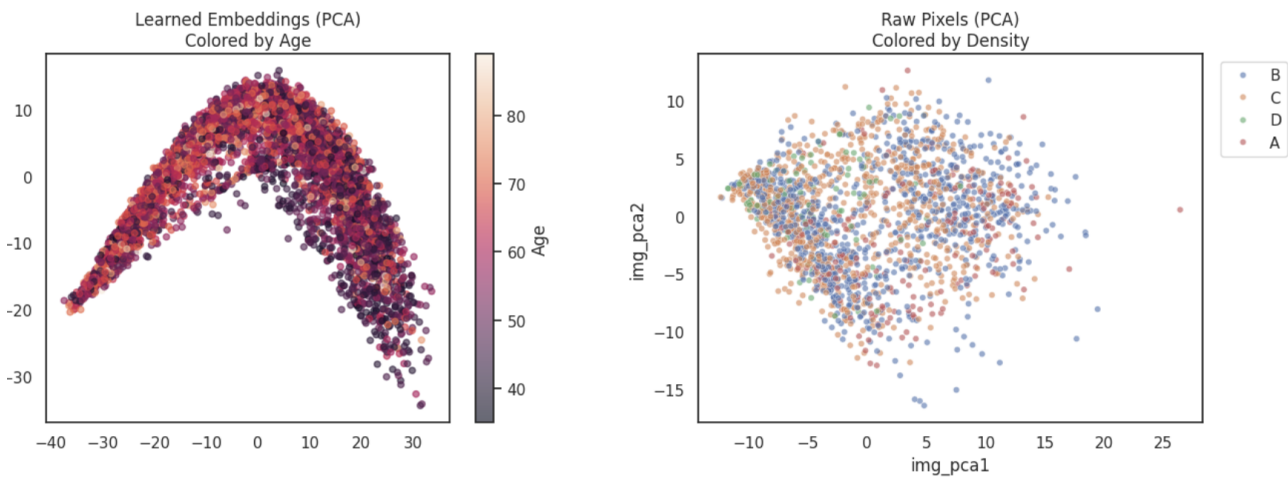


Figure 5: PCA of embeddings coloured by age (left) and PCA of flattened raw pixels coloured by density (right).

In contrast to PCA of the learned embeddings, PCA of flattened raw pixels (Figure 5) shows substantial class mixing, demonstrating that density separation emerges from learned features rather than global intensity differences.

The t-SNE projection (Figure 6) shows density-consistent local neighbourhoods. When coloured by machine ID, points remain intermixed, which suggests scanner identity does not drive embedding

geometry and is consistent with the small AUROC variation across scanners.

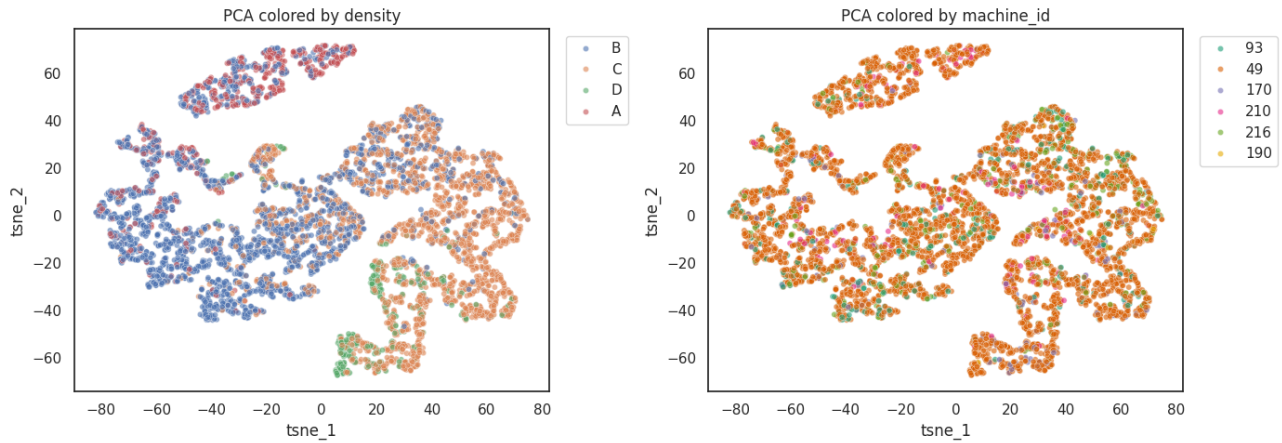


Figure 6: t-SNE projection of embeddings coloured by density (left) and machine ID (right), showing density-consistent neighbourhoods and no clear clustering by scanner.

Finally, misclassifications cluster in overlap regions between adjacent classes (Figure 7), consistent with borderline cases rather than systematic bias.

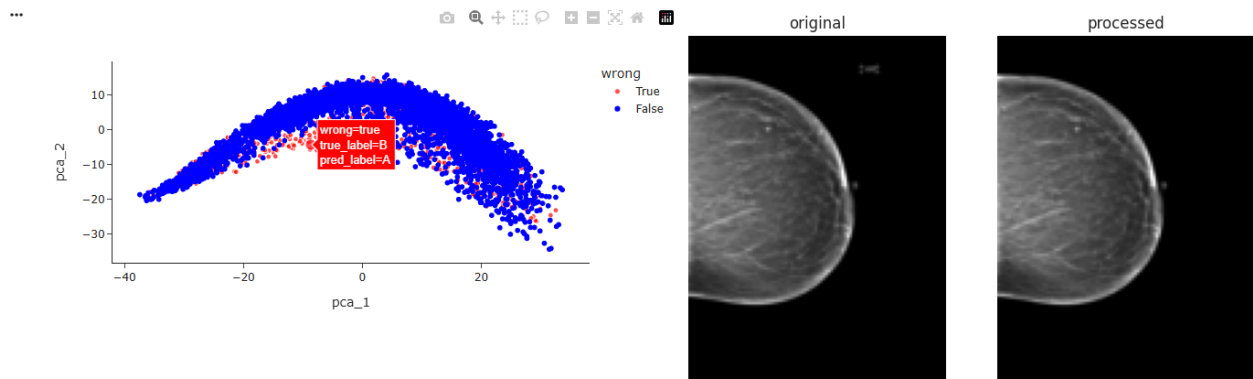


Figure 7: Embedding PCA with misclassified samples highlighted (left) and a representative borderline case (true B predicted as A) shown for inspection (right).